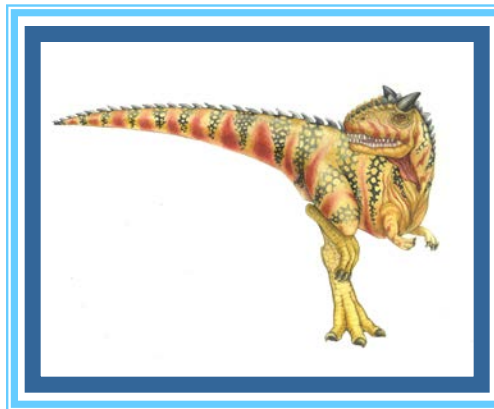


Chapter 9: Virtual Memory





Chapter 9: Virtual Memory

- Background
- Demand Paging
- Copy-on-Write
- Page Replacement
- Allocation of Frames
- Thrashing
- Memory-Mapped Files
- Allocating Kernel Memory
- Other Considerations
- Operating-System Examples





Objectives

- To describe the benefits of a virtual memory system
- To explain the concepts of demand paging, page-replacement algorithms, and allocation of page frames
- To discuss the principle of the working-set model
- To examine the relationship between shared memory and memory-mapped files
- To explore how kernel memory is managed





Background

- Code needs to be in memory to execute, but entire program rarely used
 - Error code, unusual routines, large data structures
- Entire program code not needed at same time
- Consider ability to execute partially-loaded program
 - Program no longer constrained by limits of physical memory
 - Each program takes less memory while running -> more programs run at the same time
 - ▶ Increased CPU utilization and throughput with no increase in response time or turnaround time
 - Less I/O needed to load or swap programs into memory -> each user program runs faster





Background (Cont.)

- **Virtual memory** – separation of user logical memory from physical memory
 - Only part of the program needs to be in memory for execution
 - Logical address space can therefore be much larger than physical address space
 - Allows address spaces to be shared by several processes
 - Allows for more efficient process creation
 - More programs running concurrently
 - Less I/O needed to load or swap processes





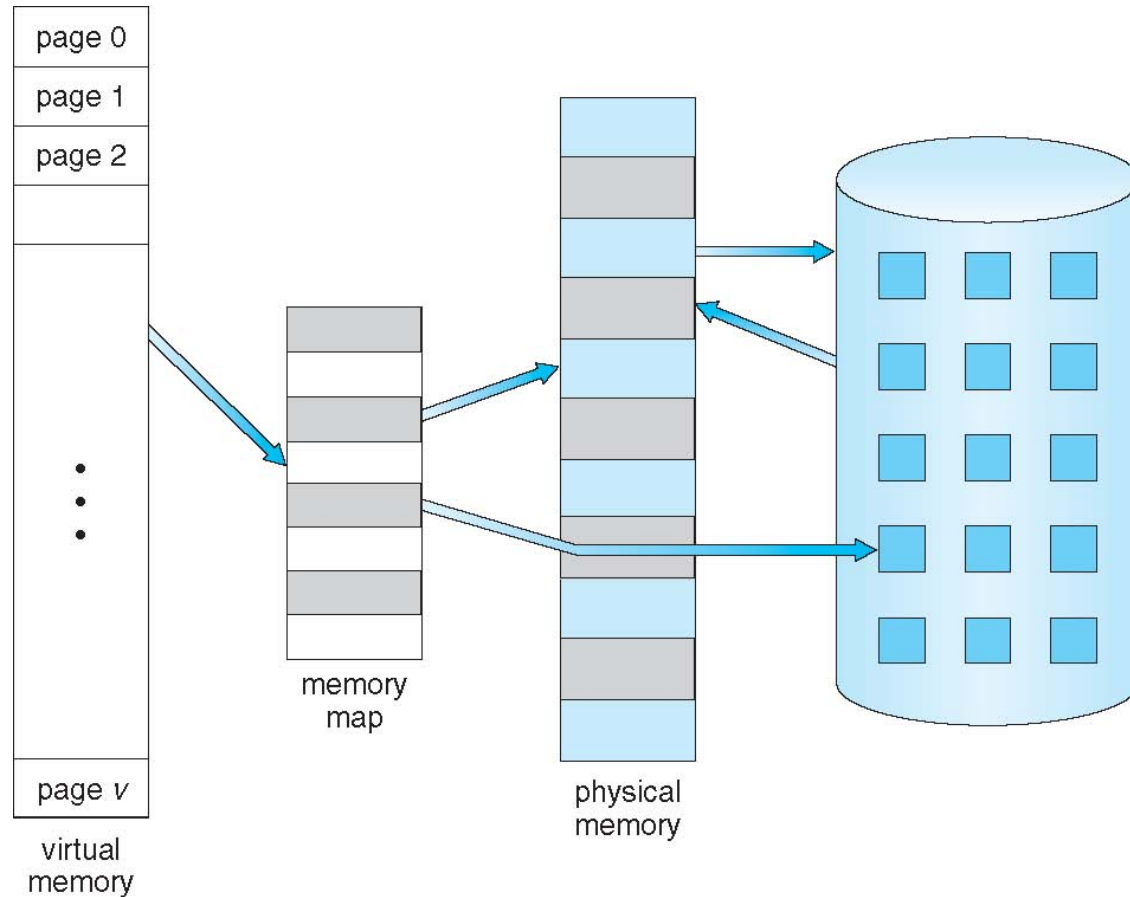
Background (Cont.)

- **Virtual address space** – logical view of how process is stored in memory
 - Usually start at address 0, contiguous addresses until end of space
 - Meanwhile, physical memory organized in page frames
 - MMU must map logical to physical
- Virtual memory can be implemented via:
 - Demand paging
 - Demand segmentation





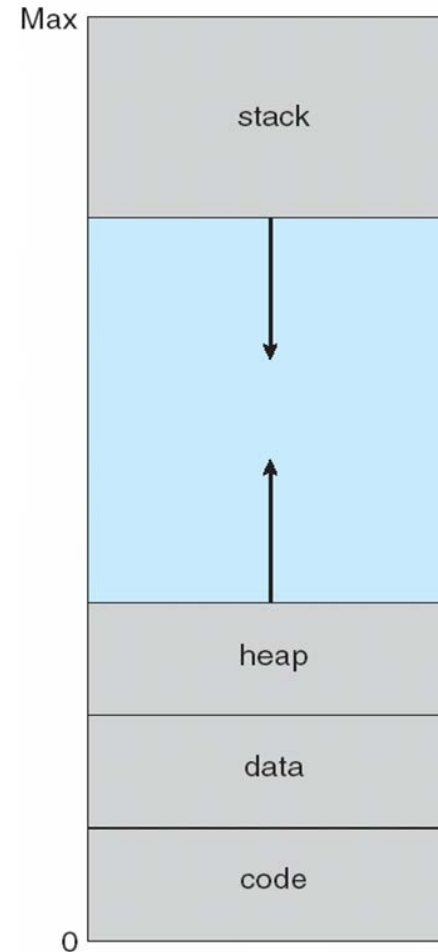
Virtual Memory That is Larger Than Physical Memory





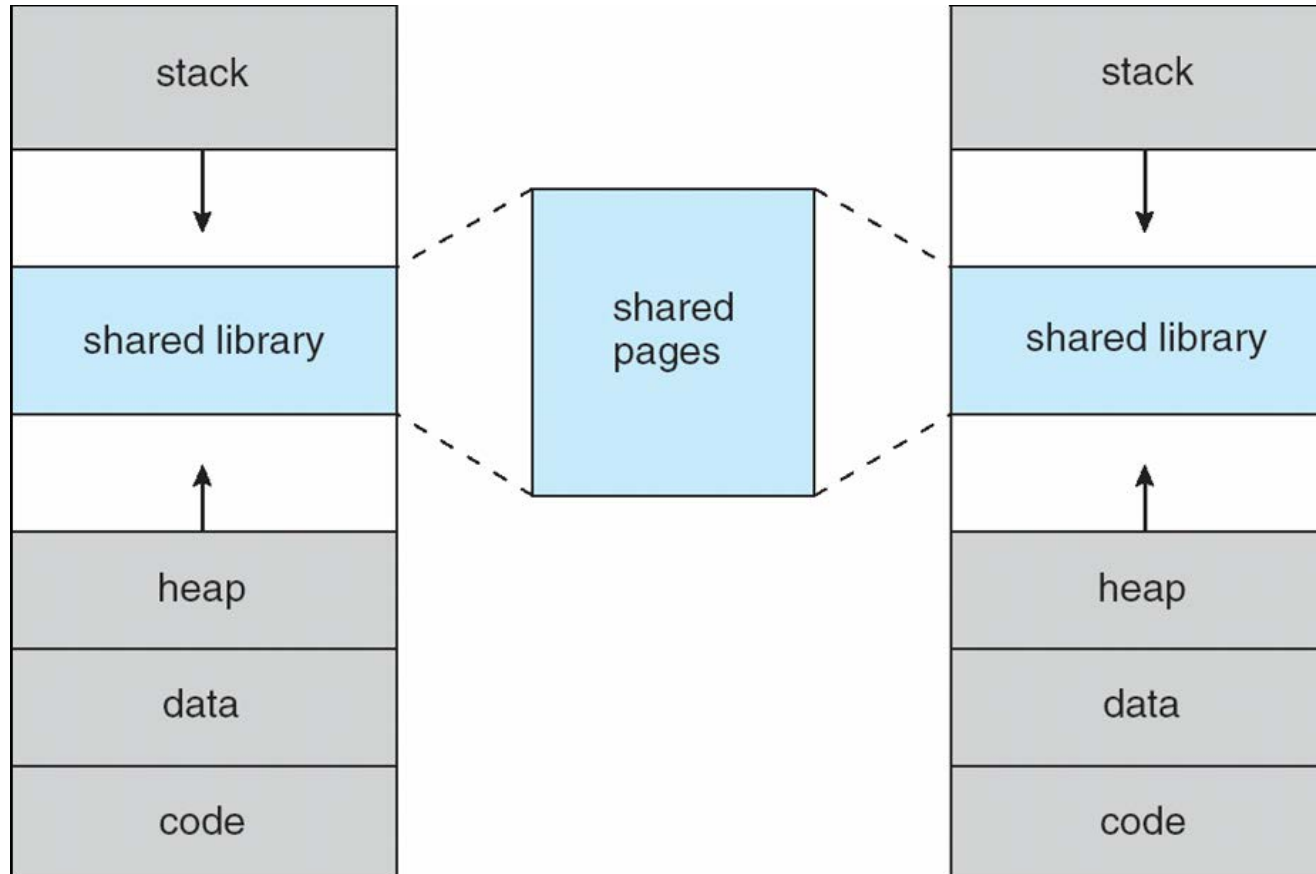
Virtual-address Space

- Usually design logical address space for stack to start at Max logical address and grow “down” while heap grows “up”
 - Maximizes address space use
 - Unused address space between the two is hole
 - ▶ No physical memory needed until heap or stack grows to a given new page
- Enables **sparse** address spaces with holes left for growth, dynamically linked libraries, etc
- System libraries shared via mapping into virtual address space
- Shared memory by mapping pages read-write into virtual address space
- Pages can be shared during `fork()`, speeding process creation





Shared Library Using Virtual Memory





Demand Paging

Demand paging is a lazy loading technique in which pages are loaded into memory only when they are needed (i.e., on demand) instead of loading the entire program at once.

1. Loading Strategy:

- **Old way:** Load the entire process into memory at once.
- **Demand paging:** Load **only the pages that are needed**, when they are needed

2. Advantages:

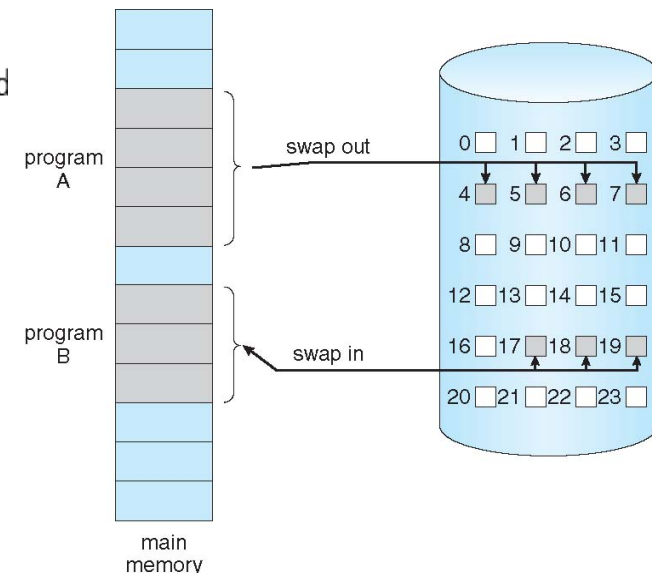
- Less I/O → avoids unnecessary loading.
- Less memory usage → faster response.
- Supports more users simultaneously.

3. How it Works:

- **Page needed** → check memory:
 - If page is **already in memory** → just use it.
 - If page is **not in memory** → page fault occurs → OS loads it from disk.
- **Invalid reference** → abort the process.

4. Pager Concept:

- **Lazy swapper:** Only swaps in a page when it is actually needed.
- Pager deals with pages instead of whole processes.





Basic Concepts

5. MMU Support:

- Demand paging requires **new memory management unit (MMU) features** to detect page faults and handle page loading **without changing program code or behavior**.

6. Key Idea:

- The program behaves as if **all pages are in memory**, but actually only some pages are loaded.





Valid-Invalid Bit

- It indicates whether the corresponding page is currently in memory.

Meaning: **Valid** (V = 1): Page is in memory → CPU can access it safely.

Invalid (V = 0): Page is not in memory or reference is illegal.

Not-in-memory: Demand paging will bring it from disk (page fault).

Illegal reference: Accessing an invalid page → OS aborts the process.

- How it works in Demand Paging:**

CPU generates a logical address → MMU checks page table.

Check the valid/invalid bit:

Valid → translate and access memory.

Invalid → raise page fault or abort (if illegal).

Frame #	valid-invalid bit
	v
	v
	v
	i
...	
	i
	i

page table





Page Table When Some Pages Are Not in Main Memory

0	A
1	B
2	C
3	D
4	E
5	F
6	G
7	H

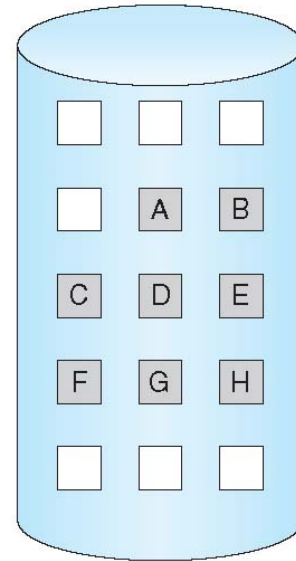
logical
memory

valid-invalid bit		
frame		
0	4	v
1		i
2	6	v
3		i
4		i
5	9	v
6		i
7		i

page table

0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

physical memory





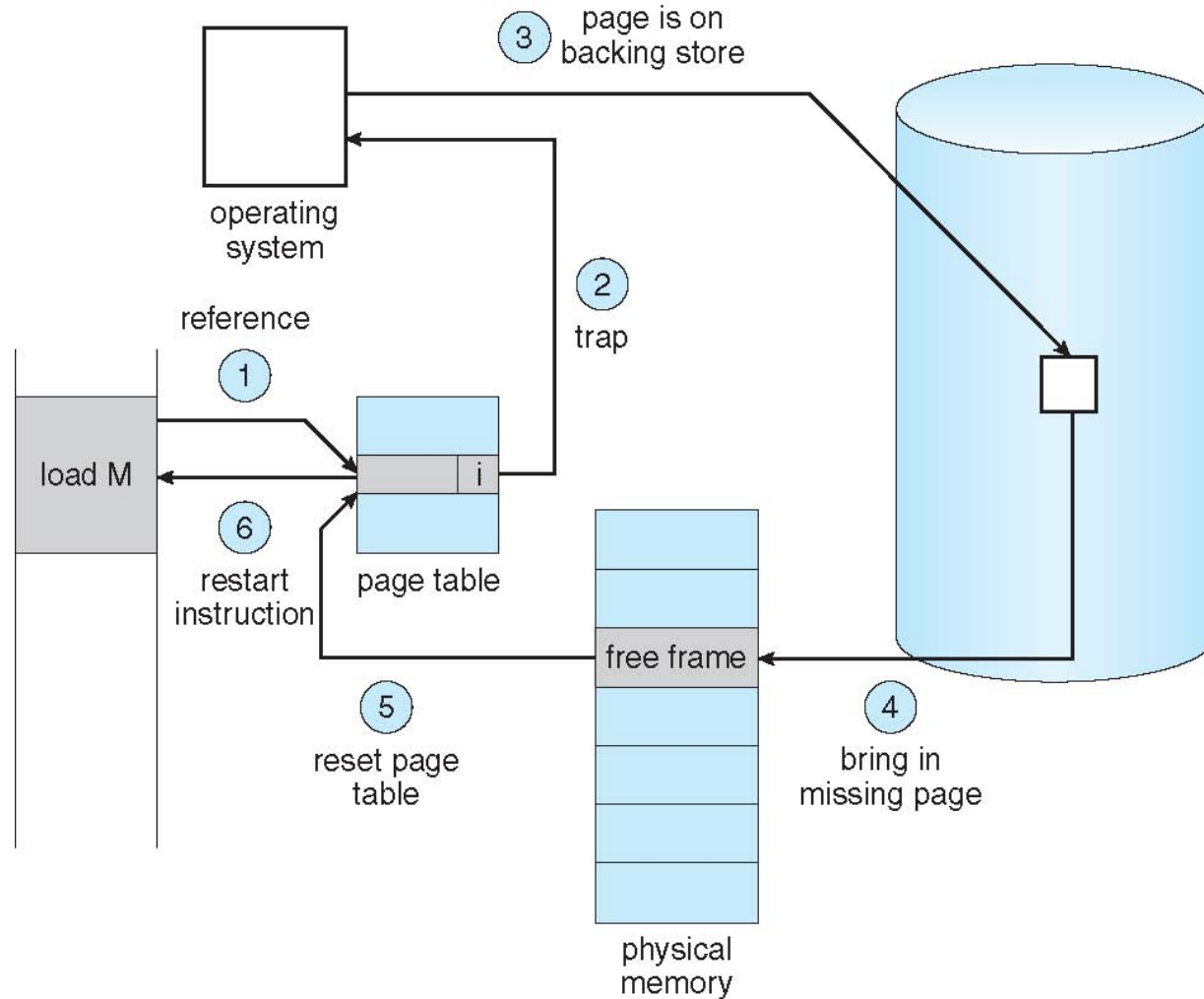
Page Fault

- A page fault occurs when a process accesses a page that is not currently in main memory, causing the operating system to load it from disk.
- 1. Operating system looks at another table to decide:
 - Invalid reference $\Rightarrow$ abort
 - Just not in memory
- 2. Find free frame
- 3. Swap page into frame via scheduled disk operation
- 4. Reset tables to indicate page now in memory Set validation bit = **V**
- 5. Restart the instruction that caused the page fault





Steps in Handling a Page Fault





Aspects of Demand Paging

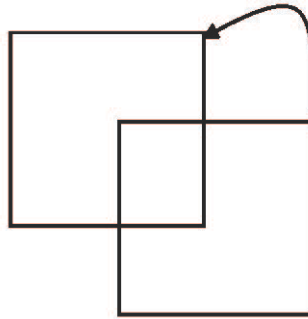
- Extreme case – start process with *no* pages in memory
 - OS sets instruction pointer to first instruction of process, non-memory-resident -> page fault
 - And for every other process pages on first access
 - **Pure demand paging**
- Actually, a given instruction could access multiple pages -> multiple page faults
 - Consider fetch and decode of instruction which adds 2 numbers from memory and stores result back to memory
 - Pain decreased because of **locality of reference**
- Hardware support needed for demand paging
 - Page table with valid / invalid bit
 - Secondary memory (swap device with **swap space**)
 - Instruction restart





Instruction Restart

- Consider an instruction that could access several different locations
 - block move



Definition:

When a **page fault occurs** while executing an instruction, the CPU may have **partially executed that instruction**. To ensure correct execution:

- The CPU **restarts the instruction from the beginning** after the page is loaded into memory.
- This ensures the instruction executes **exactly as if no page fault occurred**.





Performance of Demand Paging

■ Stages in Demand Paging (worse case)

1. Trap to the operating system
2. Save the user registers and process state
3. Determine that the interrupt was a page fault
4. Check that the page reference was legal and determine the location of the page on the disk
5. Issue a read from the disk to a free frame:
 1. Wait in a queue for this device until the read request is serviced
 2. Wait for the device seek and/or latency time
 3. Begin the transfer of the page to a free frame
6. While waiting, allocate the CPU to some other user
7. Receive an interrupt from the disk I/O subsystem (I/O completed)
8. Save the registers and process state for the other user
9. Determine that the interrupt was from the disk
10. Correct the page table and other tables to show page is now in memory
11. Wait for the CPU to be allocated to this process again
12. Restore the user registers, process state, and new page table, and then resume the interrupted instruction





Performance of Demand Paging (Cont.)

2. Effective Memory Access Time (EMAT) Formula

$$EMAT = (1 - p) \cdot ma + p \cdot pf$$

- $(1 - p) \times ma \rightarrow$ normal memory access.
- $p \times pf \rightarrow$ page fault handling time.

Even a small page fault rate drastically **increases** EMAT, because $pf \gg ma$.

■ Page Fault Rate $0 \leq p \leq 1$

- if $p = 0$ no page faults
- if $p = 1$, every reference is a fault

■ Effective Access Time (EAT)

$$\begin{aligned} EAT = & (1 - p) \times \text{memory access} \\ & + p (\text{page fault overhead} \\ & \quad + \text{swap page out} \\ & \quad + \text{swap page in}) \end{aligned}$$





Demand Paging Example

Effective Access Time (EAT)

$$EAT = (1 - p) \times (\text{memory access}) + p \times (\text{page fault overhead})$$

- **Memory access time** = time if page is in memory.
- **Page fault overhead** = interrupt service + swap out + swap in + restart.

Example Calculation:

- Memory access time = 200 ns
- Average page-fault service time = 8 ms = 8,000,000 ns

$$EAT = (1 - p) \cdot 200 + p \cdot 8,000,000 = 200 + p \cdot 7,999,800$$

- If 1 in 1,000 memory accesses causes a page fault:

$$p = 0.001$$

$$EAT = 200 + 0.001 \cdot 7,999,800 \approx 8,200 \text{ ns}$$

- This is **40× slowdown!**





Demand Paging Optimizations

- Swap space I/O faster than file system I/O even if on the same device
 - Swap allocated in larger chunks, less management needed than file system
- Copy entire process image to swap space at process load time
 - Then page in and out of swap space
 - Used in older BSD Unix
- Demand page in from program binary on disk, but discard rather than paging out when freeing frame
 - Used in Solaris and current BSD
 - Still need to write to swap space
 - ▶ Pages not associated with a file (like stack and heap) – **anonymous memory**
 - ▶ Pages modified in memory but not yet written back to the file system
- Mobile systems
 - Typically don't support swapping
 - Instead, demand page from file system and reclaim read-only pages (such as code)





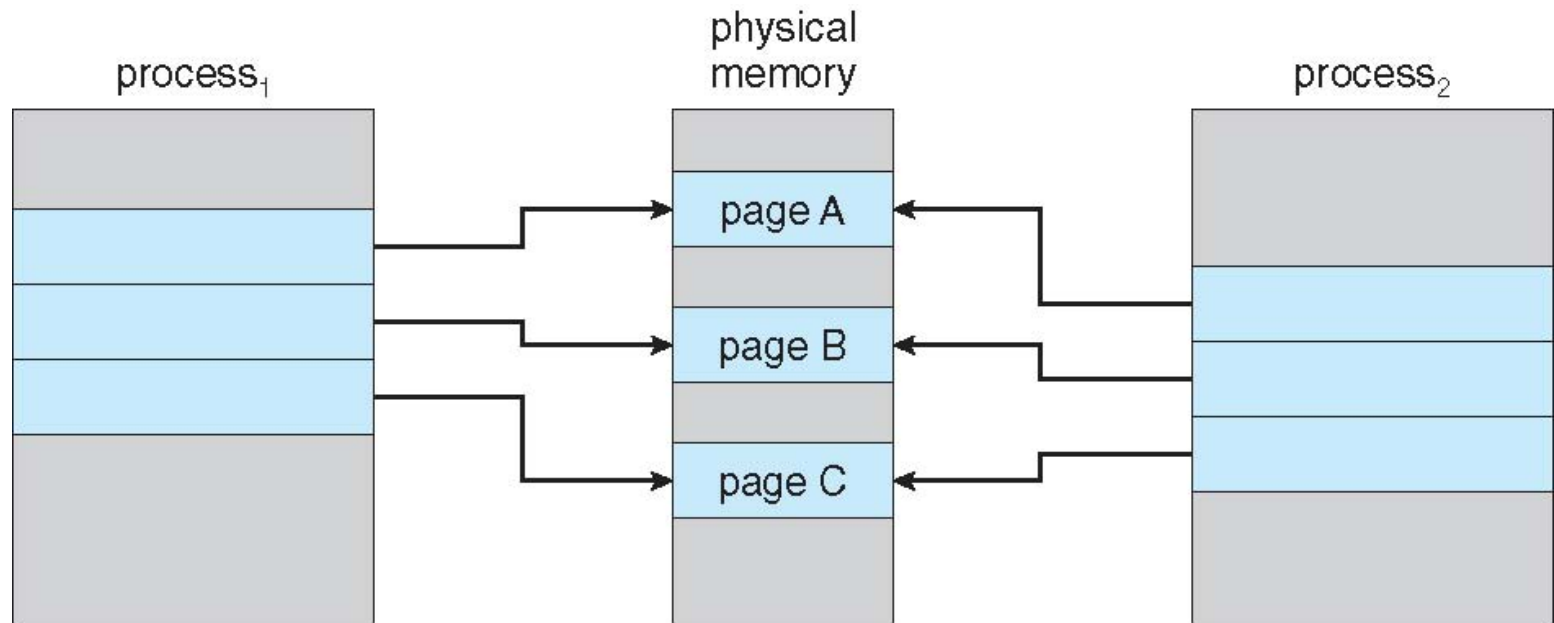
Copy-on-Write

- **Copy-on-Write** (COW) allows both parent and child processes to initially **share** the same pages in memory
 - If either process modifies a shared page, only then is the page copied
- COW allows more efficient process creation as only modified pages are copied
- In general, free pages are allocated from a **pool** of **zero-fill-on-demand** pages
 - Pool should always have free frames for fast demand page execution
 - ▶ Don't want to have to free a frame as well as other processing on page fault
 - Why zero-out a page before allocating it?
- `vfork()` variation on `fork()` system call has parent suspend and child using copy-on-write address space of parent
 - Designed to have child call `exec()`
 - Very efficient



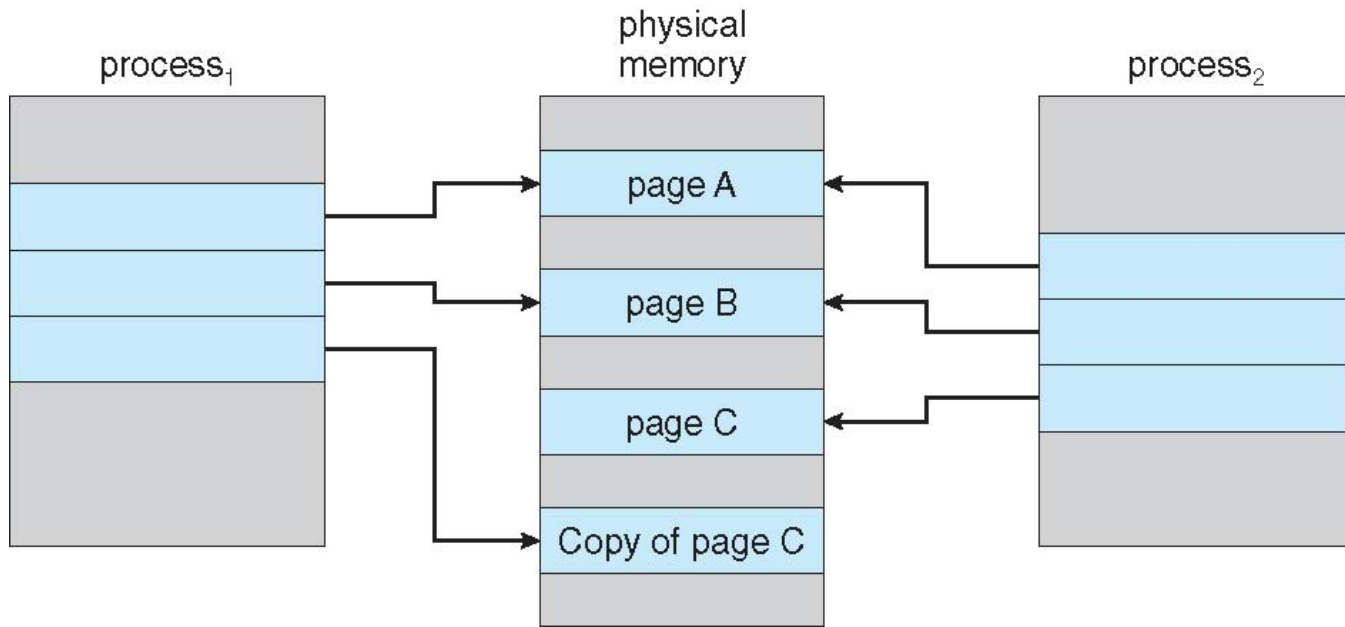


Before Process 1 Modifies Page C





After Process 1 Modifies Page C





What Happens if There is no Free Frame?

- Used up by process pages
- Also in demand from the kernel, I/O buffers, etc
- How much to allocate to each?
- Page replacement – find some page in memory, but not really in use, page it out
 - Algorithm – terminate? swap out? replace the page?
 - Performance – want an algorithm which will result in minimum number of page faults
- Same page may be brought into memory several times





Page Replacement

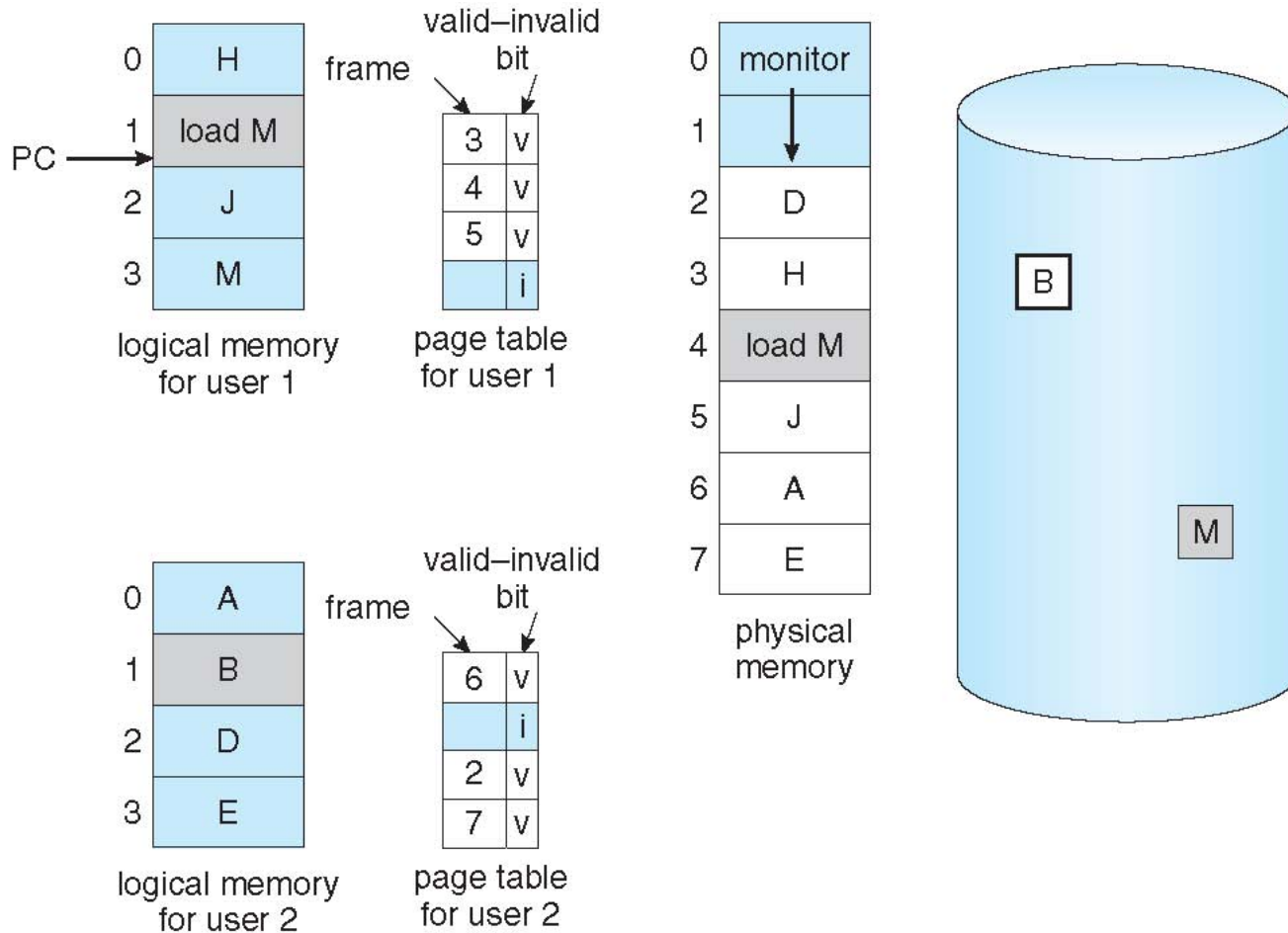
Page replacement is a memory management technique used when a page fault occurs and there is no free frame in physical memory. The operating system must choose a page in memory to remove (replace) to make room for the new page.

Page replacement helps manage memory efficiently, reduces unnecessary disk writes using the dirty bit, and allows processes to use virtual memory larger than physical memory





Need For Page Replacement





Basic Page Replacement

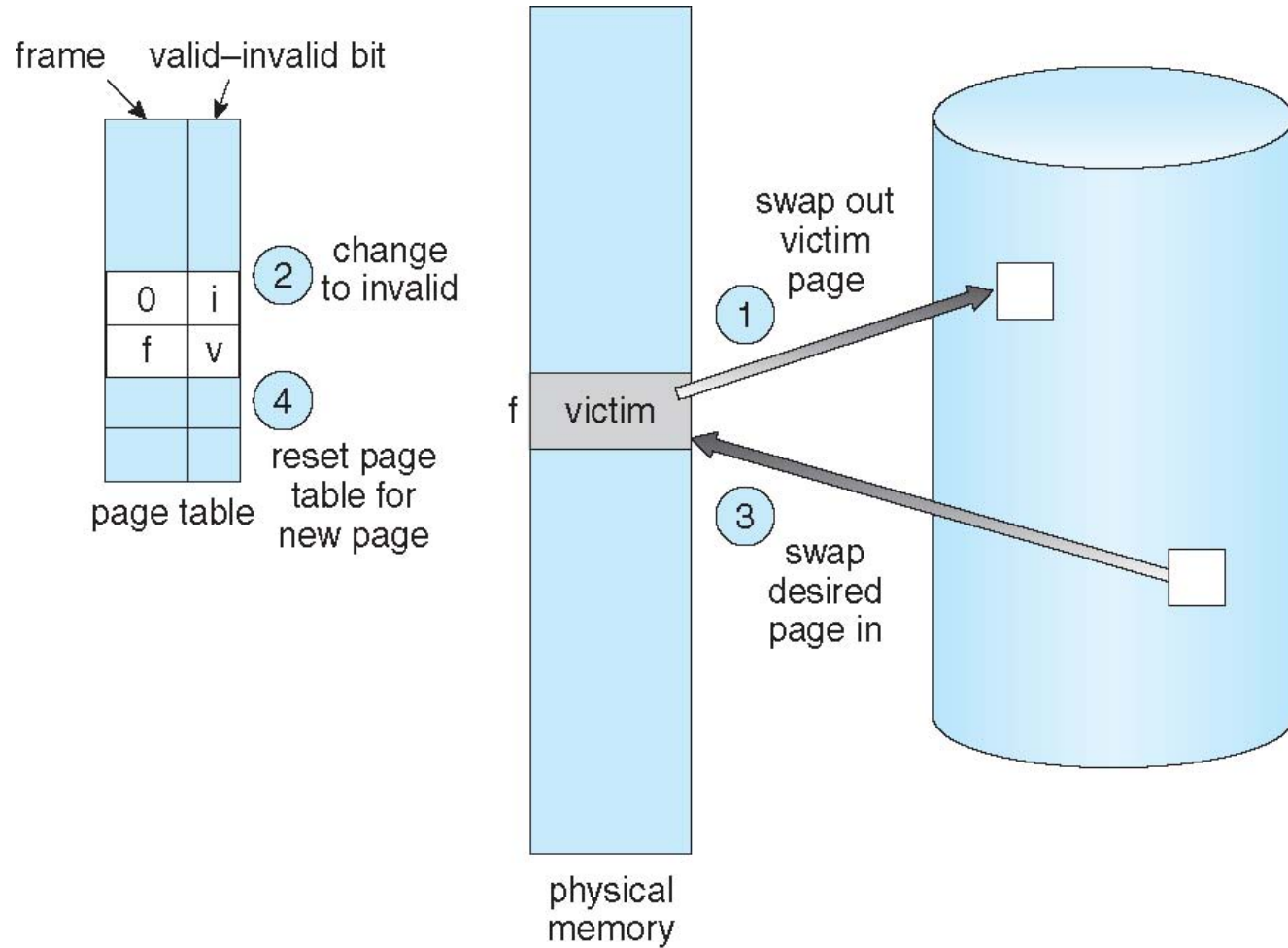
1. Find the location of the desired page on disk
2. Find a free frame:
 - If there is a free frame, use it
 - If there is no free frame, use a page replacement algorithm to select a **victim frame**
 - Write victim frame to disk if dirty
3. Bring the desired page into the (newly) free frame; update the page and frame tables
4. Continue the process by restarting the instruction that caused the trap

Note now potentially 2 page transfers for page fault – increasing EAT





Page Replacement





Page and Frame Replacement Algorithms

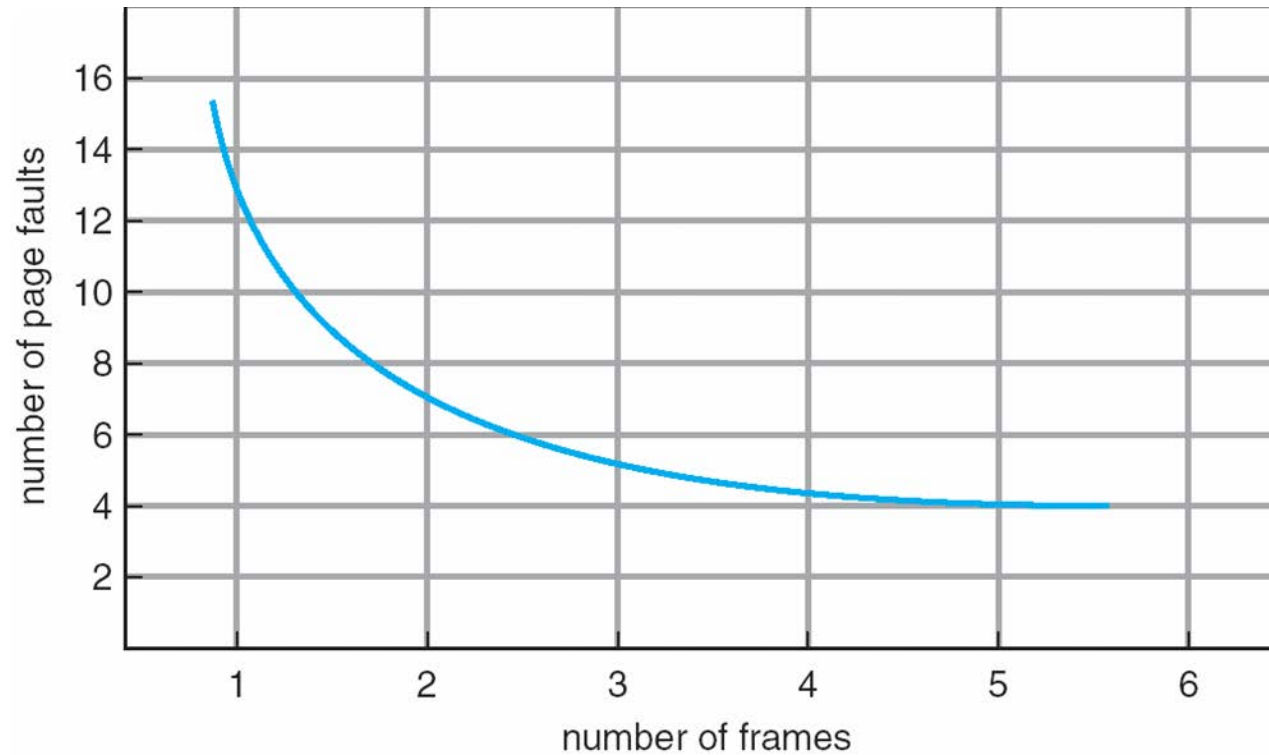
- **Frame-allocation algorithm** Determines how many frames each process gets and which frames to replace when needed.
- **Page-replacement algorithm** Determines which page in memory should be replaced on a page fault to minimize page-fault rate.
- Evaluate algorithm by running it on a particular string of memory
 1. **FIFO (First-In-First-Out)**: Replace the page that has been in memory the longest. Simple to implement, but may remove frequently used pages (**Belady's anomaly**).
 2. **Optimal (OPT)**: Replace the page that **won't be used for the longest time**. Best theoretically, but not practical since future references are unknown.
 3. **LRU (Least Recently Used)**: Replace the page that **hasn't been used for the longest time**. Approximates optimal; implemented using counters or a stack.
- In all our examples, the **reference string** of referenced page numbers is

7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1





Graph of Page Faults Versus The Number of Frames





First-In-First-Out (FIFO) Algorithm

- Reference string: **7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1**
- 3 frames (3 pages can be in memory at a time per process)

reference string

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

7	7	7	2	2	2	4	4	4	0	0	0	7	7	7
	0	0	0	3	3	3	2	2	2	1	1	1	0	0
		1	1	1	0	0	0	3	3	3	2	2	2	1

page frames

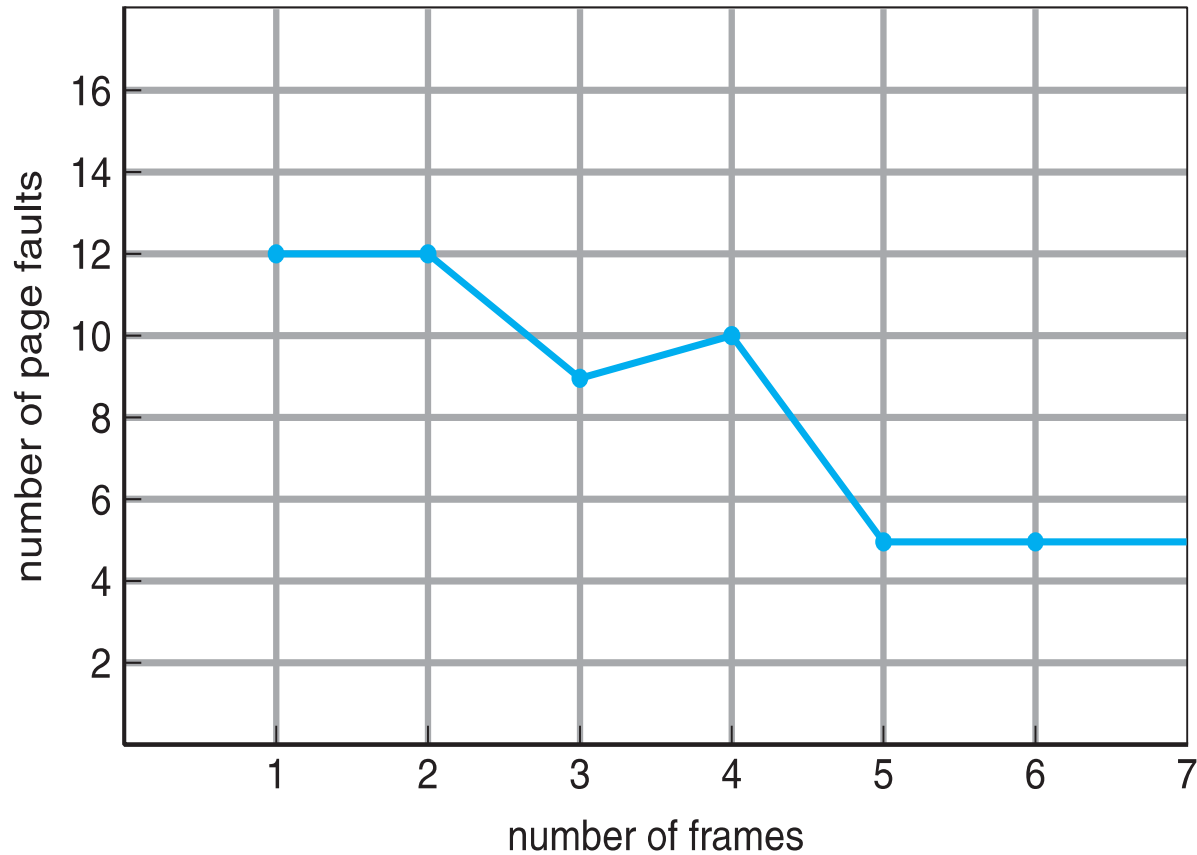
15 page faults

- Can vary by reference string: consider 1,2,3,4,1,2,5,1,2,3,4,5
 1. Maintain a **queue of pages** currently in memory.
 2. When a **page fault** occurs:
 - If there is a **free frame**, load the new page into it.
 - If all frames are **full**, **remove the page that entered memory first** (the oldest page).
 - Load the new page into the **freed frame**.





FIFO Illustrating Belady's Anomaly



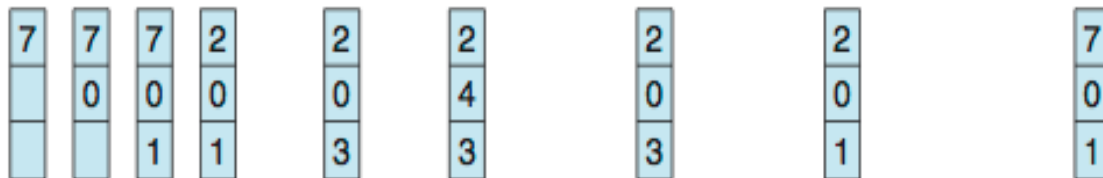


Optimal Algorithm

- When a page fault occurs and no free frame is available, the OS looks at the future page references. It chooses the page that will be needed the farthest in the future. That page is replaced, and the new page is loaded into memory.
- Used for measuring how well your algorithm performs

reference string

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1



page frames





Least Recently Used (LRU) Algorithm

- When a page fault occurs and memory is full, the OS checks which page was used least recently. That page is replaced by the new page.
- Associate time of last use with each page

reference string

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

7	7	7	2		2		4	4	4	0			1		1		1		
	0	0	0		0		0	0	3	3			3		0		0		
		1	1		3		3	2	2	2			2		2		7		

page frames

- 12 faults – better than FIFO but worse than OPT
- Generally good algorithm and frequently used
- But how to implement?





LRU Algorithm (Cont.)

1. Counter Implementation

- Each page entry has a **counter**.
- Every time a page is **accessed**, the current **clock or timestamp** is copied into the counter.
- When a page needs to be replaced, **find the page with the smallest counter value** → this is the **least recently used page**.
- **Disadvantage:** You have to **search through all counters**, which can be slow.

2. Stack Implementation

- Maintain a **stack of page numbers** (often double-linked).
- When a page is **referenced**, move it to the **top of the stack**.
- The page at the **bottom** of the stack is the **least recently used** → easy to replace.
- **Disadvantage:** Updating the stack requires **changing multiple pointers** (about 6), so each access is more expensive.
- **Advantage:** No need to search for a replacement page.

- **Both LRU and OPT are examples of stack algorithms. They do not suffer from Belady's anomaly (unlike FIFO)**





Use Of A Stack to Record Most Recent Page References

reference string

4 7 0 7 1 0 1 2 1 2 7 1 2

2
1
0
7
4

stack
before
a

7
2
1
0
4

stack
after
b

↑
a

↑
b





LRU Approximation Algorithms

- LRU needs special hardware and still slow
- **Reference bit**
 - With each page associate a bit, initially = 0
 - When page is referenced bit set to 1
 - Replace any with reference bit = 0 (if one exists)
 - ▶ We do not know the order, however
- **Second-chance algorithm**
 - Generally FIFO, plus hardware-provided reference bit
 - **Clock** replacement
 - If page to be replaced has
 - ▶ Reference bit = 0 -> replace it
 - ▶ reference bit = 1 then:
 - set reference bit 0, leave page in memory
 - replace next page, subject to same rules





Second-Chance (clock) Page-Replacement Algorithm



next
victim



circular queue of pages

(a)





Enhanced Second-Chance Algorithm

- Improve algorithm by using reference bit and modify bit (if available) in concert
- Take ordered pair (reference, modify)
 1. (0, 0) neither recently used nor modified – best page to replace
 2. (0, 1) not recently used but modified – not quite as good, must write out before replacement
 3. (1, 0) recently used but clean – probably will be used again soon
 4. (1, 1) recently used and modified – probably will be used again soon and need to write out before replacement
- When page replacement called for, use the clock scheme but use the four classes replace page in lowest non-empty class
 - Might need to search circular queue several times





Counting Algorithms

- Keep a counter of the number of references that have been made to each page
 - Not common
- **Least Frequently Used (LFU) Algorithm**: replaces page with smallest count
- **Most Frequently Used (MFU) Algorithm**: based on the argument that the page with the smallest count was probably just brought in and has yet to be used





Page-Buffering Algorithms

- Keep a pool of free frames, always
 - Then frame available when needed, not found at fault time
 - Read page into free frame and select victim to evict and add to free pool
 - When convenient, evict victim
- Possibly, keep list of modified pages
 - When backing store otherwise idle, write pages there and set to non-dirty
- Possibly, keep free frame contents intact and note what is in them
 - If referenced again before reused, no need to load contents again from disk
 - Generally useful to reduce penalty if wrong victim frame selected





Applications and Page Replacement

- All of these algorithms have OS guessing about future page access
- Some applications have better knowledge – i.e. databases
- Memory intensive applications can cause double buffering
 - OS keeps copy of page in memory as I/O buffer
 - Application keeps page in memory for its own work
- Operating system can given direct access to the disk, getting out of the way of the applications
 - **Raw disk** mode
- Bypasses buffering, locking, etc





Allocation of Frames

- Each process needs ***minimum*** number of frames
- Example: IBM 370 – 6 pages to handle SS MOVE instruction:
 - instruction is 6 bytes, might span 2 pages
 - 2 pages to handle *from*
 - 2 pages to handle *to*
- ***Maximum*** of course is total frames in the system
- Two major allocation schemes
 - fixed allocation
 - priority allocation
- Many variations





Fixed Allocation

- Equal allocation – For example, if there are 100 frames (after allocating frames for the OS) and 5 processes, give each process 20 frames
 - Keep some as free frame buffer pool
- Proportional allocation – Allocate according to the size of process
 - Dynamic as degree of multiprogramming, process sizes change
 - s_i = size of process p_i
 - $S = \sum s_i$
 - m = total number of frames
 - a_i = allocation for $p_i = \frac{s_i}{S} \times m$

$$m = 64$$

$$s_1 = 10$$

$$s_2 = 127$$

$$a_1 = \frac{10}{137} \times 62 \approx 4$$

$$a_2 = \frac{127}{137} \times 62 \approx 57$$





Priority Allocation

- Use a proportional allocation scheme using priorities rather than size
- If process P_i generates a page fault,
 - select for replacement one of its frames
 - select for replacement a frame from a process with lower priority number





Global vs. Local Allocation

- **Global replacement** – process selects a replacement frame from the set of all frames; one process can take a frame from another
 - But then process execution time can vary greatly
 - But greater throughput so more common

- **Local replacement** – each process selects from only its own set of allocated frames
 - More consistent per-process performance
 - But possibly underutilized memory





Non-Uniform Memory Access

■ NUMA is a computer memory design used in multiprocessor systems where the time it takes for a processor to access memory depends on the memory's location. Some memory is closer (faster) to the processor, while other memory is farther away (slower).

■ **Optimal performance requires:**

1. Allocating memory close to the CPU where the thread runs.
2. Modifying the scheduler to run the thread on the same CPU/board whenever possible.

● NUMA improves performance by allowing each processor to access its local memory quickly. It is useful for large-scale multiprocessor systems with shared memory





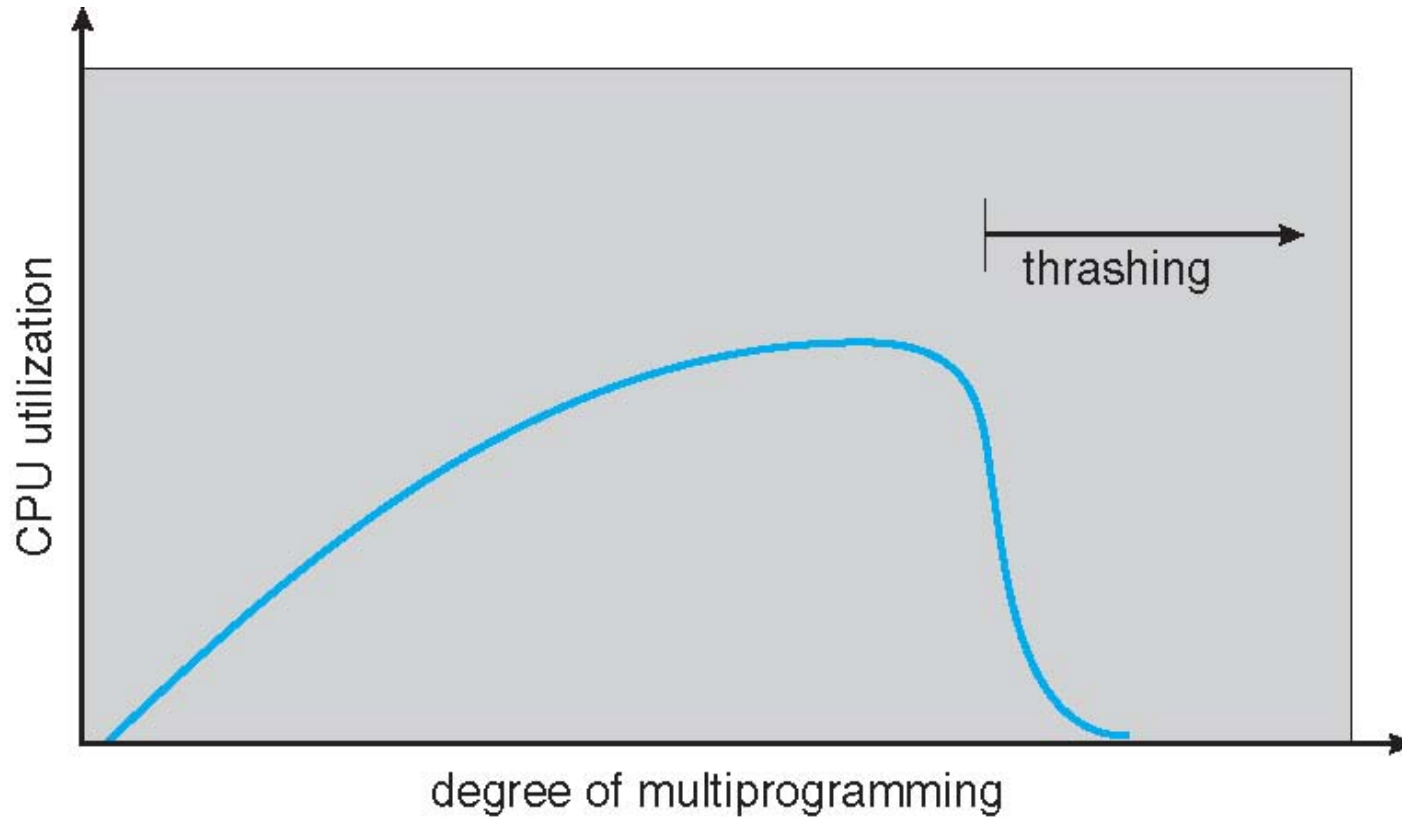
Thrashing

- Thrashing occurs when a process keeps swapping pages due to insufficient memory, leading to low CPU utilization and poor system performance
 - This leads to:
 - ▶ Low CPU utilization
 - ▶ Operating system thinking that it needs to increase the degree of multiprogramming
 - ▶ Another process added to the system
- **Thrashing** $\equiv$ Thrashing happens when a process is constantly busy swapping pages in and out instead of executing instructions.





Thrashing (Cont.)





Demand Paging and Thrashing

- Why does demand paging work?

Locality model

- Process migrates from one locality to another
- Localities may overlap

- Why does thrashing occur?

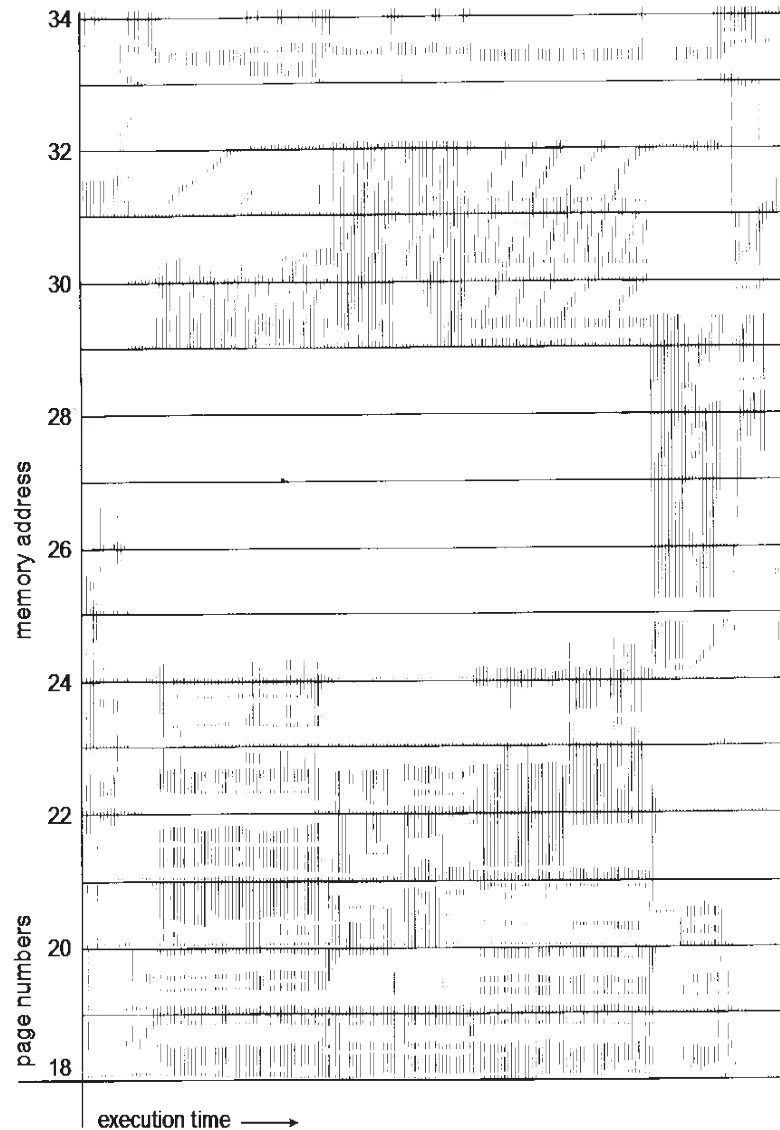
Σ size of locality > total memory size

- Limit effects by using local or priority page replacement





Locality In A Memory-Reference Pattern



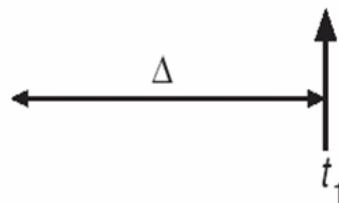


Working-Set Model

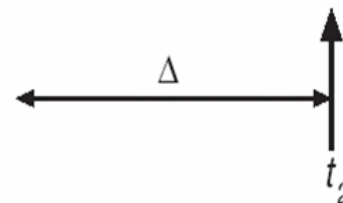
- $\Delta \equiv$ working-set window $\equiv$ a fixed number of page references
Example: 10,000 instructions
- WSS_i (working set of Process P_i) =
total number of pages referenced in the most recent Δ (varies in time)
 - if Δ too small will not encompass entire locality
 - if Δ too large will encompass several localities
 - if $\Delta = \infty \Rightarrow$ will encompass entire program
- $D = \sum WSS_i \equiv$ total demand frames
 - Approximation of locality
- if $D > m \Rightarrow$ Thrashing
- Policy if $D > m$, then suspend or swap out one of the processes

page reference table

... 2 6 1 5 7 7 7 7 5 1 6 2 3 4 1 2 3 4 4 4 3 4 3 4 4 4 4 1 3 2 3 4 4 4 3 4 4 4 ...



$$WS(t_1) = \{1, 2, 5, 6, 7\}$$



$$WS(t_2) = \{3, 4\}$$





Keeping Track of the Working Set

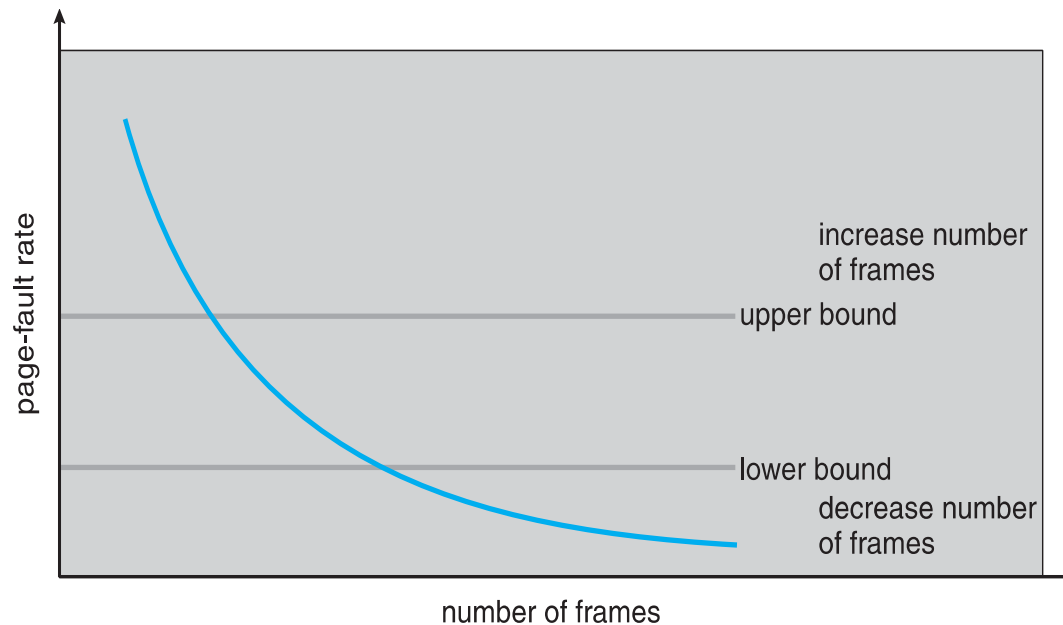
- Approximate with interval timer + a reference bit
- Example: $\Delta = 10,000$
 - Timer interrupts after every 5000 time units
 - Keep in memory 2 bits for each page
 - Whenever a timer interrupts copy and sets the values of all reference bits to 0
 - If one of the bits in memory = 1 $\Rightarrow$ page in working set
- Why is this not completely accurate?
- Improvement = 10 bits and interrupt every 1000 time units





Page-Fault Frequency

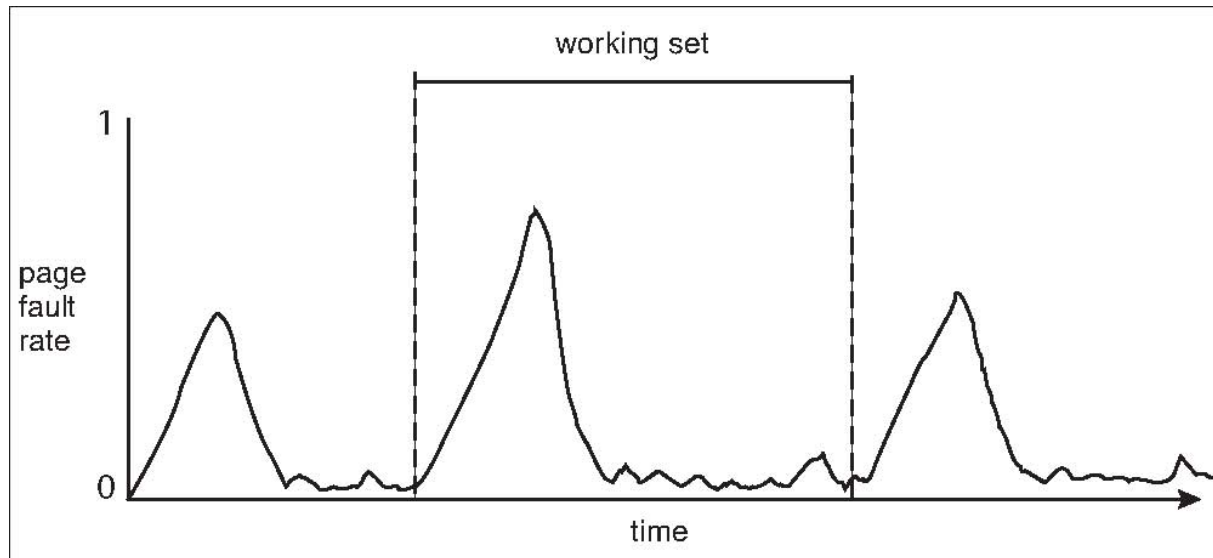
- More direct approach than WSS
- Establish “acceptable” **page-fault frequency (PFF)** rate and use local replacement policy
 - If actual rate too low, process loses frame
 - If actual rate too high, process gains frame





Working Sets and Page Fault Rates

- Direct relationship between working set of a process and its page-fault rate
- Working set changes over time
- Peaks and valleys over time





Memory-Mapped Files

- Memory-mapped file I/O allows file I/O to be treated as routine memory access by **mapping** a disk block to a page in memory
- A file is initially read using demand paging
 - A page-sized portion of the file is read from the file system into a physical page
 - Subsequent reads/writes to/from the file are treated as ordinary memory accesses
- Simplifies and speeds file access by driving file I/O through memory rather than `read()` and `write()` system calls
- Also allows several processes to map the same file allowing the pages in memory to be shared
- But when does written data make it to disk?
 - Periodically and / or at `file close()` time
 - For example, when the pager scans for dirty pages





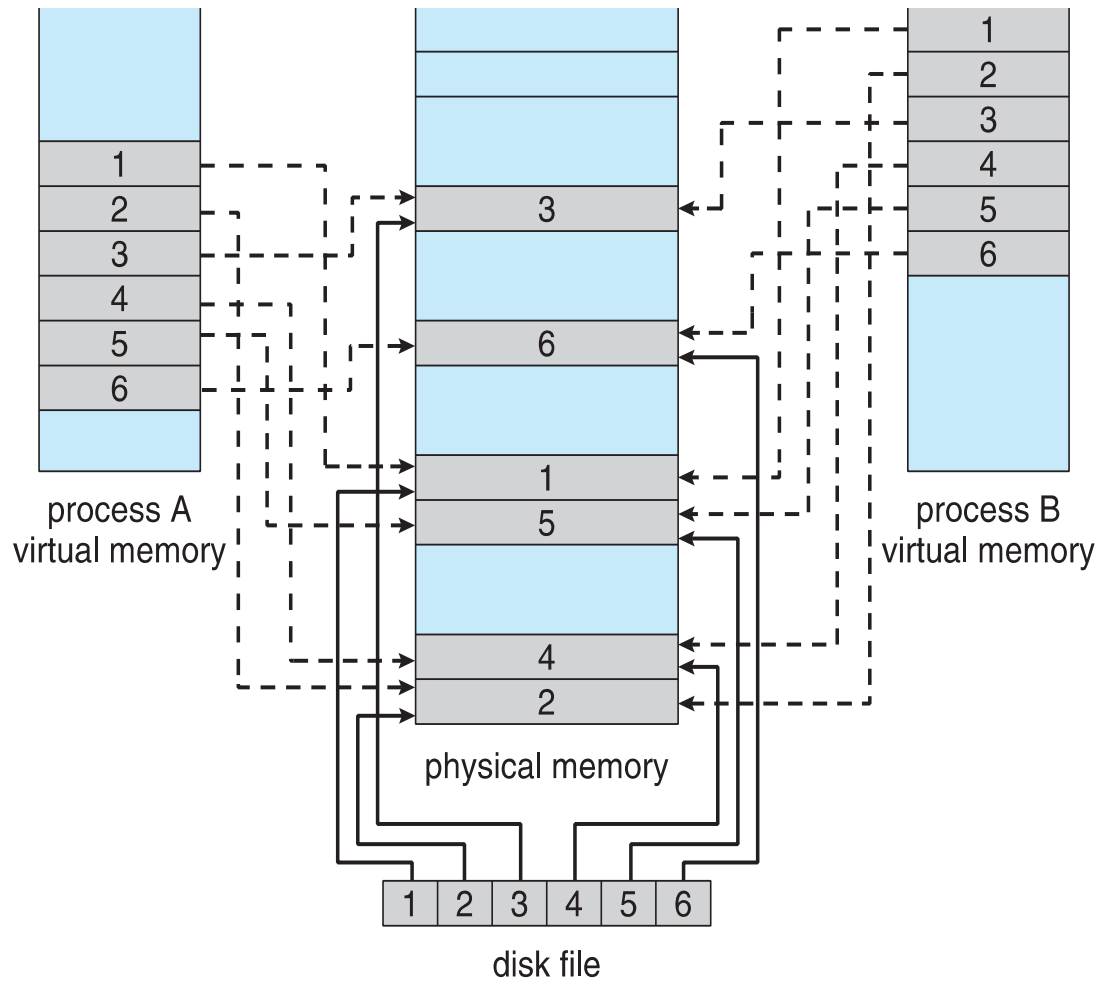
Memory-Mapped File Technique for all I/O

- Some OSes use memory mapped files for standard I/O
- Process can explicitly request memory mapping a file via `mmap()` system call
 - Now file mapped into process address space
- For standard I/O (`open()`, `read()`, `write()`, `close()`), `mmap` anyway
 - But map file into kernel address space
 - Process still does `read()` and `write()`
 - ▶ Copies data to and from kernel space and user space
 - Uses efficient memory management subsystem
 - ▶ Avoids needing separate subsystem
- COW can be used for read/write non-shared pages
- Memory mapped files can be used for shared memory (although again via separate system calls)



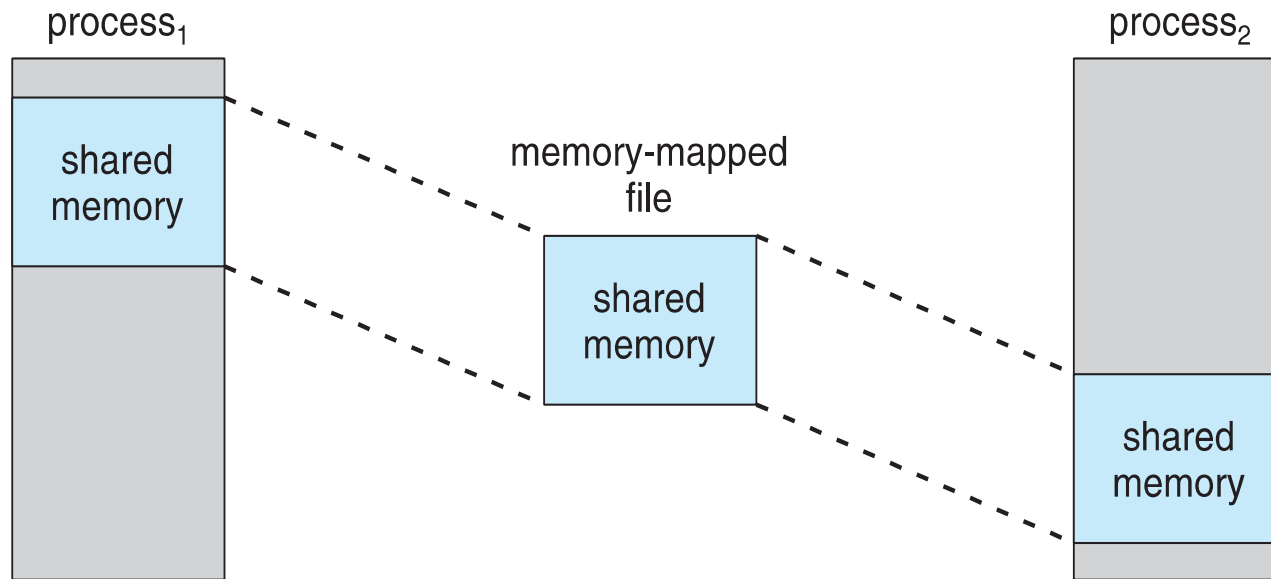


Memory Mapped Files





Shared Memory via Memory-Mapped I/O





Shared Memory in Windows API

- First create a **file mapping** for file to be mapped
 - Then establish a view of the mapped file in process's virtual address space
- Consider producer / consumer
 - Producer create shared-memory object using memory mapping features
 - Open file via `CreateFile()`, returning a `HANDLE`
 - Create mapping via `CreateFileMapping()` creating a **named shared-memory object**
 - Create view via `MapViewOfFile()`
- Sample code in Textbook





Allocating Kernel Memory

- Treated differently from user memory
- Often allocated from a free-memory pool
 - Kernel requests memory for structures of varying sizes
 - Some kernel memory needs to be contiguous
 - ▶ I.e. for device I/O





Buddy System

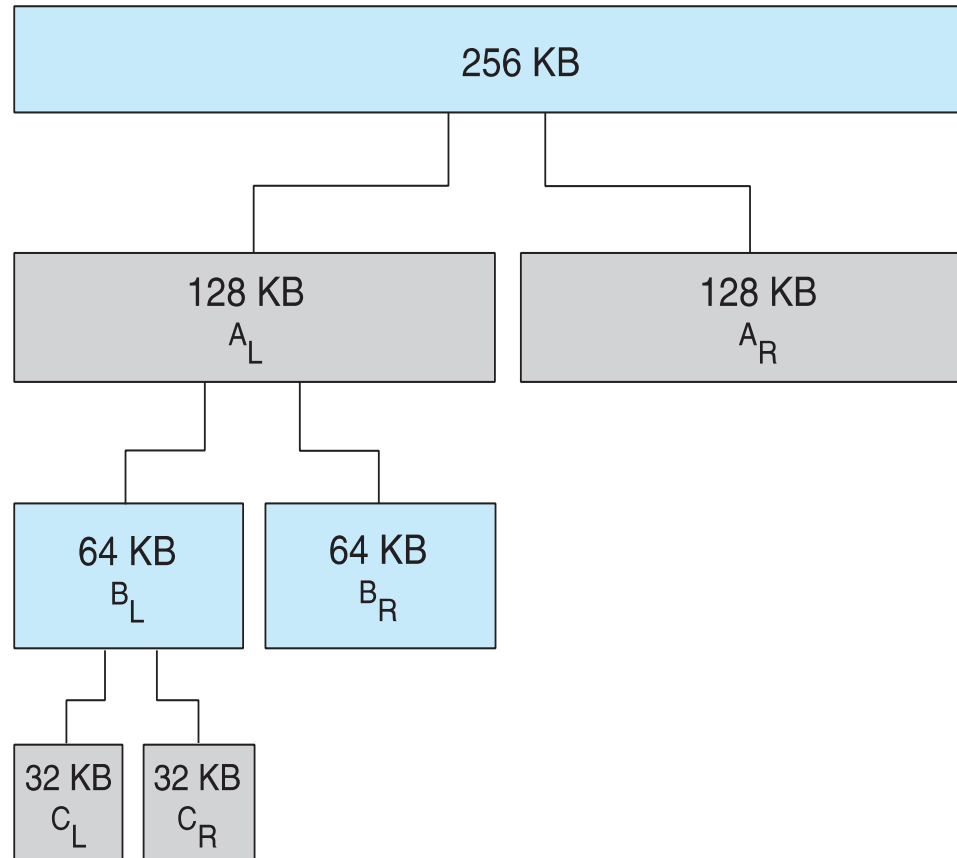
- Allocates memory from fixed-size segment consisting of physically-contiguous pages
- Memory allocated using **power-of-2 allocator**
 - Satisfies requests in units sized as power of 2
 - Request rounded up to next highest power of 2
 - When smaller allocation needed than is available, current chunk split into two buddies of next-lower power of 2
 - ▶ Continue until appropriate sized chunk available
- For example, assume 256KB chunk available, kernel requests 21KB
 - Split into A_L and A_R of 128KB each
 - ▶ One further divided into B_L and B_R of 64KB
 - One further into C_L and C_R of 32KB each – one used to satisfy request
- Advantage – quickly **coalesce** unused chunks into larger chunk
- Disadvantage - fragmentation





Buddy System Allocator

physically contiguous pages





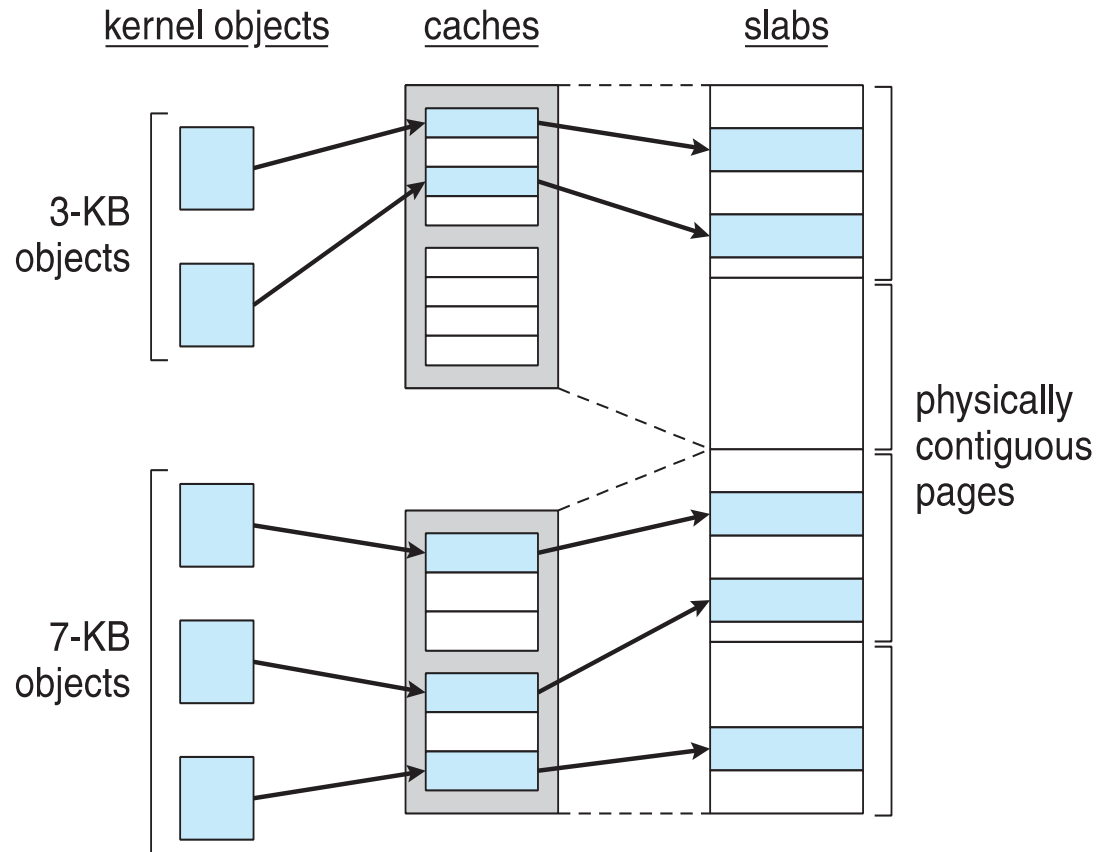
Slab Allocator

- Alternate strategy
- **Slab** is one or more physically contiguous pages
- **Cache** consists of one or more slabs
- Single cache for each unique kernel data structure
 - Each cache filled with **objects** – instantiations of the data structure
- When cache created, filled with objects marked as **free**
- When structures stored, objects marked as **used**
- If slab is full of used objects, next object allocated from empty slab
 - If no empty slabs, new slab allocated
- Benefits include no fragmentation, fast memory request satisfaction





Slab Allocation





Slab Allocator in Linux

- For example process descriptor is of type struct `task_struct`
- Approx 1.7KB of memory
- New task -> allocate new struct from cache
 - Will use existing free struct `task_struct`
- Slab can be in three possible states
 1. Full – all used
 2. Empty – all free
 3. Partial – mix of free and used
- Upon request, slab allocator
 1. Uses free struct in partial slab
 2. If none, takes one from empty slab
 3. If no empty slab, create new empty





Slab Allocator in Linux (Cont.)

- Slab started in Solaris, now wide-spread for both kernel mode and user memory in various OSes
- Linux 2.2 had SLAB, now has both SLOB and SLUB allocators
 - SLOB for systems with limited memory
 - ▶ Simple List of Blocks – maintains 3 list objects for small, medium, large objects
 - SLUB is performance-optimized SLAB removes per-CPU queues, metadata stored in page structure





Other Considerations -- Prepaging

■ Prepaging

- To reduce the large number of page faults that occurs at process startup
- Prepage all or some of the pages a process will need, before they are referenced
- But if prepaged pages are unused, I/O and memory was wasted
- Assume s pages are prepaged and α of the pages is used
 - ▶ Is cost of $s * \alpha$ save pages faults $>$ or $<$ than the cost of prepaging
 $s * (1 - \alpha)$ unnecessary pages?
 - ▶ α near zero $\Rightarrow$ prepaging loses





Other Issues – Page Size

- Sometimes OS designers have a choice
 - Especially if running on custom-built CPU
- Page size selection must take into consideration:
 - Fragmentation
 - Page table size
 - **Resolution**
 - I/O overhead
 - Number of page faults
 - Locality
 - TLB size and effectiveness
- Always power of 2, usually in the range 2^{12} (4,096 bytes) to 2^{22} (4,194,304 bytes)
- On average, growing over time





Other Issues – TLB Reach

- TLB Reach - The amount of memory accessible from the TLB
- $\text{TLB Reach} = (\text{TLB Size}) \times (\text{Page Size})$
- Ideally, the working set of each process is stored in the TLB
 - Otherwise there is a high degree of page faults
- Increase the Page Size
 - This may lead to an increase in fragmentation as not all applications require a large page size
- Provide Multiple Page Sizes
 - This allows applications that require larger page sizes the opportunity to use them without an increase in fragmentation





Other Issues – Program Structure

■ Program structure

- `int[128,128] data;`
- Each row is stored in one page
- Program 1

```
for (j = 0; j < 128; j++)  
    for (i = 0; i < 128; i++)  
        data[i,j] = 0;
```

128 x 128 = 16,384 page faults

- Program 2

```
for (i = 0; i < 128; i++)  
    for (j = 0; j < 128; j++)  
        data[i,j] = 0;
```

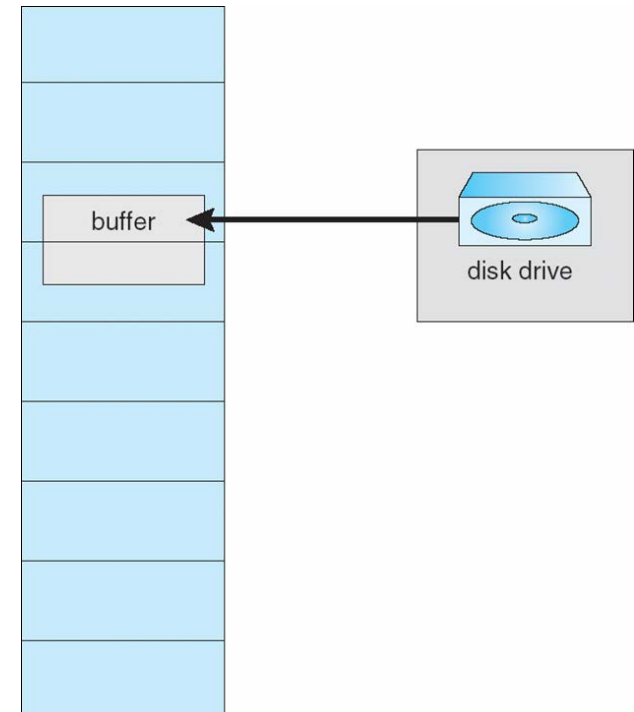
128 page faults





Other Issues – I/O interlock

- **I/O Interlock** – Pages must sometimes be locked into memory
- Consider I/O - Pages that are used for copying a file from a device must be locked from being selected for eviction by a page replacement algorithm
- **Pinning** of pages to lock into memory





Operating System Examples

- Windows
- Solaris





Windows

- Uses demand paging with **clustering**. Clustering brings in pages surrounding the faulting page
- Processes are assigned **working set minimum** and **working set maximum**
- Working set minimum is the minimum number of pages the process is guaranteed to have in memory
- A process may be assigned as many pages up to its working set maximum
- When the amount of free memory in the system falls below a threshold, **automatic working set trimming** is performed to restore the amount of free memory
- Working set trimming removes pages from processes that have pages in excess of their working set minimum





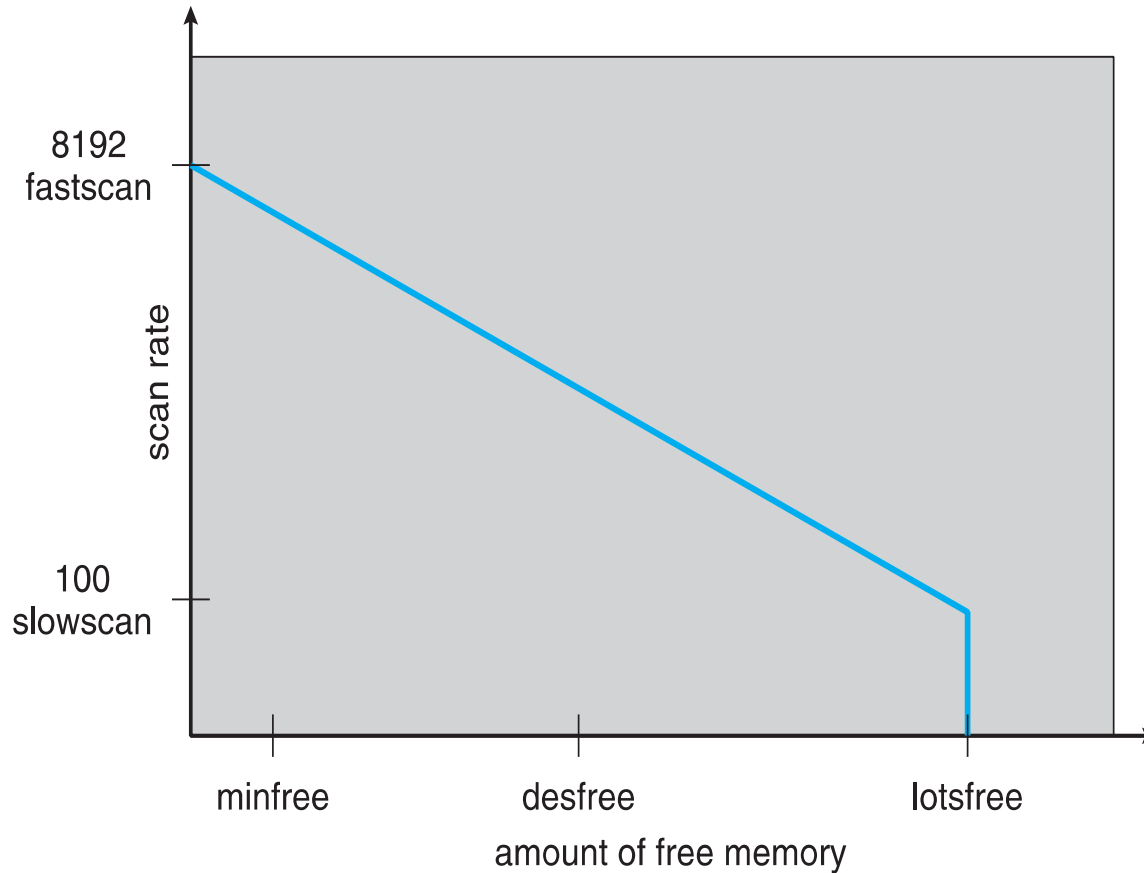
Solaris

- Maintains a list of free pages to assign faulting processes
- **Lotsfree** – threshold parameter (amount of free memory) to begin paging
- **Desfree** – threshold parameter to increasing paging
- **Minfree** – threshold parameter to being swapping
- Paging is performed by **pageout** process
- **Pageout** scans pages using modified clock algorithm
- **Scanrate** is the rate at which pages are scanned. This ranges from **slowscan** to **fastscan**
- **Pageout** is called more frequently depending upon the amount of free memory available
- **Priority paging** gives priority to process code pages





Solaris 2 Page Scanner



End of Chapter 9

